

# Quality Control Standards — Nexus Semiconductor Fab 3

Hillsboro, OR · Document ID: NXS-QC-001 · Revision 2 · Effective 20 Mar 2026

|                |                                     |                |                      |
|----------------|-------------------------------------|----------------|----------------------|
| Document ID    | NXS-QC-001                          | Revision       | 2                    |
| Plant          | Nexus Semiconductor Fab 3           | Effective Date | 20 Mar 2026          |
| Document Owner | V. Acharya, Yield & Quality Manager | Classification | Controlled — Quality |

## 1. Purpose & Scope

This standard defines wafer-level quality control at Nexus Semiconductor Fab 3. It covers in-line metrology, defect inspection, parametric test, wafer sort, and the SPC/FDC framework. Aligned with SEMI E133 (process control system) and SEMI E10.

## 2. Product Specifications

| Product          | Key Attributes                     | Specification Reference                                          |
|------------------|------------------------------------|------------------------------------------------------------------|
| Lithography CD   | Critical dimension uniformity      | 3-sigma $\leq 2.0$ nm at gate layer; OPC verification mandatory. |
| Etch profile     | Sidewall angle, depth              | $89^\circ \pm 0.5^\circ$ ; depth uniformity 3-sigma $\leq 2\%$ . |
| Defect density   | Adders post-lithography, post-etch | $\leq 0.05$ defects/cm <sup>2</sup> for critical layers.         |
| Parametric (PCM) | Vt, drive current, leakage         | Yield-relevant parameters within process window.                 |

## 3. Procedures

### 3.1 SPC and FDC

Every CPP under SPC. Western Electric rules drive alerts. Fault Detection and Classification (FDC) on tool sensor data feeds an upstream alert engine before SPC trips.

### 3.2 Lot Disposition

Out-of-control lots placed on hold until disposition by Process Integration. Disposition codes mapped to engineering and Quality through the MES.

### 3.3 Yield Review

Daily yield review across lithography, etch, and EOL test. Top three defect signatures reviewed at the weekly cross-functional Yield Council.

### 3.4 Customer-Specific Requirements

Automotive (AEC-Q100) and Aerospace customers require additional Cpk floors and Part Average Testing. These flow down into the MES recipe per customer ID.

## 4. Out-of-Specification Handling

Any OOS result triggers an immediate Hold on the affected lot, a documented investigation, and where applicable, a CAPA. Disposition decisions (Release, Rework, Reject) require QA approval. The investigation is closed in the eQMS with full traceability per the site's records retention policy.

## 5. Roles & Responsibilities

Quality Assurance (V. Acharya, Yield & Quality Manager) owns lot release. Operations executes per the recipe and reports deviations. Maintenance supports calibration of QC-critical instruments per the linked maintenance manual (NXS-MNT-001). The site safety procedure (NXS-SAF-001) applies to all QC laboratory activity.

## 6. References

- SEMI E10, E35, E133
- AEC-Q100 (for automotive lots)
- JEDEC reliability standards
- Internal: SPC-NXS-GATE-01, YIELD-NXS-COUNCIL-WI

## 7. Revision History

| Rev | Date        | Author     | Summary                                  |
|-----|-------------|------------|------------------------------------------|
| 1   | 18 Sep 2024 | V. Acharya | Initial issue.                           |
| 2   | 20 Mar 2026 | V. Acharya | Updated specifications and SPC criteria. |

## Approvals

| Role                    | Name                                | Signature | Date        |
|-------------------------|-------------------------------------|-----------|-------------|
| Quality Assurance       | V. Acharya, Yield & Quality Manager | _____     | 20 Mar 2026 |
| Operations / Site Lead  | Q. Sandberg, Fab GM                 | _____     | 20 Mar 2026 |
| Regulatory / Compliance | Site Compliance Officer             | _____     | 20 Mar 2026 |

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